

Discipline of Electrical and Electronic Engineering
EE502 Project Laboratory: Report Marking Sheet

Introduction (5%)

This section should define the project and place its importance in context.

- Clear explanation of laparoscopic training challenges – *2 marks*
- Definition of project objectives and scope – *2 marks*
- Relevance to bioinstrumentation / clinical safety – *1 mark*

Design Strategy and Methodology (25%)

This section should explain design decisions and how the system was developed.

- System architecture and overall design approach – *5 marks*
- Justification of sensor selection and measured variables – *5 marks*
- Description of data processing (filtering, calibration, mapping to motion) – *5 marks*
- Design of AR feedback strategy (how feedback guides the user) – *5 marks*
- Consideration of enhancements or extended functionality – *5 marks*
(e.g., *grip sensing, slip detection, workspace limits, tool-tip estimation*)

Design Implementation (20%)

Assessment of how the system was physically and technically realised.

- Hardware integration (sensors, Arduino/Pi, interfacing) – *5 marks*
- Software implementation (data acquisition + AR integration) – *5 marks*
- Mechanical design and construction – *10 marks*
 - Includes use and/or modification of 3D printed files
 - Credit for design improvements, ergonomics, or robustness

Testing and Verification (30%)

This section should assess how well the system meets the design requirements.

- Description of testing methodology – *5 marks*
- Demonstration of real-time feedback behaviour (warnings, thresholds, responsiveness) – *5 marks*
- Evaluation of sensor performance (accuracy, noise, stability) – *5 marks*
- Quantitative performance metrics (at least 3) (e.g., time, smoothness, deviation, depth control) – *10 marks*
- Evidence that system distinguishes good vs poor performance – *5 marks*

Conclusion (10%)

A reflective assessment of the project outcome.

- Summary of system performance and achievements – *3 marks*
- Critical evaluation of limitations and challenges – *3 marks*
- Discussion of improvements and future work (*including possible enhancements beyond core requirements*) – *4 marks*

Presentation (10%)

Assessment of report quality and clarity.

- Structure and logical organisation – *3 marks*
- Clarity of technical explanation – *3 marks*
- Quality of figures/diagrams (system, results, AR visuals) – *2 marks*
- Grammar, formatting, and professionalism – *2 marks*